Assignment 3

Statistical Inference 2026–27

November 5, 2026

Discussion among yourselves is permitted and encouraged. AI usage is permitted.

The focus of this assignment is communicating statistical evidence. Submit mathematical arguments, reproducible R code, and a written report of at most 5 pages (single-spaced, 11 pt font, 1 in margin), excluding code. Both datasets are included in R's `datasets` package.

Problem 1: Briefing the admissions committee (10 points)

A university's admissions committee is reviewing historical records after a newspaper reported that female applicants had a lower overall admission rate. One committee member says:

> "If the overall difference is statistically significant, the data prove that admissions decisions discriminated against women."

Another replies:

> "Once we account for departments, any overall gap can be dismissed."

You are asked to prepare an evidence-based briefing that evaluates both statements.

Use `UCBAdmissions`, which cross-tabulates admission outcome, recorded gender, and department for 4,526 applicants to the six largest UC Berkeley departments in 1973. Work with the recorded categories `Female` and `Male`, and departments A–F (see also the data documentation).

a. Simpson's Paradox (3 points) Let r_{gd} be the admission probability for gender $g \in \{F, M\}$ in department d, and let w_{gd} be the proportion of that gender's applicants applying to department d. Thus $r_g = \sum_d w_{gd} r_{gd}$. Show that

$$r_F - r_M = \sum_d w_{Fd}(r_{Fd} - r_{Md}) + \sum_d (w_{Fd} - w_{Md}) r_{Md}.$$

Use this identity to explain how a pooled difference can have the opposite sign to every within-department difference. Is such a reversal possible if the department weights are identical?

b. Prepare the initial briefing. (3 points) Collapse over departments and construct the gender-by-admission table. Report admission rates, the female-minus-male difference in percentage points, and an appropriate test of equal admission probabilities. State its null and justify its calibration.

Write a **note for the committee chair** explaining what the pooled analysis establishes and assessing the first committee member's claim. Make clear what the p-value measures in this historical-data exercise.

c. Investigate the department explanation. (4 points) Calculate admission rates and female-minus-male differences within all six departments. Make one labeled figure comparing the within-department and pooled differences. Then standardize both genders' admission rates using the same department weights: each department's share of all applicants in the dataset. Compare the standardized and pooled differences.

Write a **recommendation to the committee**. Explain what changed and why, evaluate the second member's claim, and identify one additional type of information needed before drawing conclusions about admissions fairness. Explain whose department mix your standardized comparison represents.

Problem 2: Communicating an ozone regression (15 points)

An environmental monitoring manager asks your team to study how ozone levels vary with weather. A preliminary report claims:

> "Temperature significantly predicts ozone. Our model therefore shows how much ozone would fall if we lowered temperature, and its confidence interval tells us the range for an individual day's ozone reading."

The manager wants an analysis that separates what the model can describe from what it can predict, and an explanation suitable for colleagues without statistical training.

Use `airquality`, containing 153 daily records from New York, May–September 1973. The response `Ozone` is mean ozone concentration in ppb from 1–3 p.m. at Roosevelt Island. The predictors are maximum daily temperature `Temp` in degrees Fahrenheit, wind speed `Wind` in mph, and solar radiation `Solar.R` in Langleys. The measurements were collected at different locations and times of day. (see also the data documentation).

a. Variance vs variance. (3 points) Consider the idealized fixed-design model

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim N_n(0, \sigma^2 I_n),$$

with X of full column rank. At a fixed predictor vector x_0, let

$$\mu_0 = x_0^T\beta, \qquad \widehat{\mu}_0 = x_0^T\widehat{\beta}, \qquad Y_0 = x_0^T\beta + \varepsilon_0,$$

where $\varepsilon_0 \sim N(0, \sigma^2)$ independently of the fitted sample. Using $\mathrm{Cov}(\widehat{\beta} \mid X) = \sigma^2(X^TX)^{-1}$, derive $\mathrm{Var}(\widehat{\mu}_0 - \mu_0 \mid X)$ and $\mathrm{Var}(Y_0 - \widehat{\mu}_0 \mid X)$. What do these quantities represent?

b. Assess the temperature claim using multiple regression. (4 points) Report the number of retained and excluded days. Fit a simple linear regression of ozone on temperature, followed by a multiple regression on temperature, wind, and solar radiation, with an intercept and no interactions. Compare the estimated temperature coefficients and their 95% confidence intervals. For the multiple regression, test whether the temperature coefficient is zero and express the estimated association for a 10°F higher temperature.

Write an **explanation for the manager** describing how the temperature association changes after adjustment and what "holding wind and solar radiation fixed" means. Explain why using the same observations matters, and whether either model supports the claim that lowering temperature would reduce ozone.

c. Communicate uncertainty for a specified day. (4 points) Using the multiple-regression model, consider a new day from the same hypothetical setting with temperature 80°F, wind speed 10 mph, and solar radiation 200 Langleys. Report the fitted mean ozone, a 95% confidence interval for the mean, and a 95% prediction interval for one new day's reading. Check whether these predictor values are within the observed ranges, and explain why that check alone does not establish that their joint combination is well supported.

Write a **memo for the manager** explaining which interval addresses uncertainty about one day's reading and why it is wider than the interval for the mean. State the assumptions behind its nominal coverage and identify one practical obstacle to using this model for a day-ahead forecast.

d. Evaluate whether the model is adequate. (4 points) Investigate the multiple-regression model using a residual-versus-fitted plot and one additional diagnostic of your choice. State what each diagnostic examines and describe what you actually observe. Explain the implications for the coefficient inference in part (b) or the prediction interval in part (c). Do not treat a diagnostic plot as proof that an assumption holds or fails, and do not remove observations solely because they have large residuals.

Write a **final recommendation for the manager** explaining whether you would use the model for describing associations, predicting individual days, both, or neither. Support your recommendation with specific diagnostic findings, identify one limitation arising from missing observations or the sequence of daily measurements, and propose one concrete next step before operational use.